SECTION 5: Electric Potential Energy & Equipotential Surfaces (Q91–110)

91. Two point charges $q_1 = 2\mu C$ and $q_2 = 3\mu C$ are 1 m apart. Potential energy of system:
    A) 54 J
    B) 54 μJ
    C) 0.054 J
    D) 0
92. Three charges, each 1 μC, at vertices of equilateral triangle, side 1 m. Total potential energy:
    A) 3 k μJ
    B) k μJ
    C) 6 k μJ
    D) Zero
93. Two equal charges separated by 1 m. Work done in bringing a third charge midway:
    A) $k q^2 / 0.5$
    B) $k q^2$
    C) $2 k q^2$
    D) 0
94. Equipotential surfaces for a point charge:
    A) Spheres centered at charge
    B) Planes
    C) Cylinders
    D) Cones
95. Equipotential surfaces for dipole:
    A) Complex curved surfaces
    B) Spheres
    C) Planes
    D) Cylinders
96. Work done in moving charge along equipotential:
    A) Zero
    B) Maximum
    C) Depends on path
    D) Non-zero constant
97. Potential energy of dipole in uniform E, θ = 0°:
    A) -pE
    B) Zero
    C) +pE
    D) Maximum
98. Potential at point on axis of dipole at distance r >> d:
    A) $k p / r^2$
    B) Zero
    C) $k p / r^3$
    D) $k p / d^2$
99. Potential at point on equatorial plane of dipole:
    A) Zero
    B) $k p / r^2$
    C) $k p / r^3$
    D) $k p / d^2$

100. Two dipoles aligned along same line, torque on one due to other:
A) Depends on separation and orientation
B) Zero
C) Maximum always
D) Independent of distance

101. Electric field lines:
A) Perpendicular to equipotential surfaces
B) Parallel to equipotentials
C) Random
D) Circular

102. A charge moves from point at potential 100 V to 50 V. Work done by field on charge q = 2 μC:
A) 1×10^{-4} J
B) 1×10^{-4} μJ
C) 100 J
D) 0

103. Two charges +q and -q separated by 2a. Potential at midpoint:
A) Zero
B) kq / a
C) -kq / a
D) 2 kq / a

104. Potential energy of system of three charges: +q at origin, +q at x = a, -q at x = 2a:
A) $k q^2 / a$
B) $k q^2 / 2a$
C) $-k q^2 / 2a$
D) 0

105. Equipotential for line charge:
A) Cylindrical surfaces around line
B) Planes perpendicular to line
C) Spheres
D) Cones

106. Electric field at point on equipotential surface:
A) Perpendicular to surface
B) Tangent to surface
C) Zero always
D) Depends on charge

107. Work done in moving charge along equipotential:
A) Zero
B) qV
C) Maximum
D) Negative

108. Two charges of same sign brought closer: potential energy:
A) Increases
B) Decreases
C) Remains same
D) Zero

109. Charge q placed near dipole along axial line, force:
A) Attractive or repulsive depending on q sign
B) Always attractive

C) Always repulsive
D) Zero

110. Potential energy stored in system of four charges at corners of square:
A) $6 k q^2 / a$
B) $4 k q^2 / a$
C) $2 k q^2 / a$
D) Zero

SECTION 6: Advanced Capacitor & Dipole Problems (Q111–130)

111. Three capacitors, 2 μF, 3 μF, 6 μF, in series. Equivalent capacitance:
A) 1 μF
B) 2 μF
C) 3 μF
D) 6 μF

112. Same capacitors in parallel: equivalent capacitance:
A) 11 μF
B) 6 μF
C) 3 μF
D) 1 μF

113. Capacitor charged to V, disconnected, dielectric K inserted. New voltage:
A) V / K
B) K·V
C) V
D) Zero

114. Two capacitors in series, one 3 μF, one 6 μF, across 12 V battery. Voltage across 3 μF:
A) 8 V
B) 4 V
C) 6 V
D) 12 V

115. Energy stored in above series combination:
A) 36 μJ
B) 72 μJ
C) 54 μJ
D) 12 μJ

116. Dipole of moment 2 μC·m in E = 5000 N/C. Torque if θ = 30°:
A) 5000 μN·m
B) 5000 N·m
C) 1730 N·m
D) 1000 N·m

117. Potential at midpoint of dipole of ±q separated by 2a:
A) Zero
B) k q / a
C) -k q / a
D) k q / 2a

118. Electric field on axial line of dipole at distance r >> d:
A) $2 k p / r^3$

B) $k\,p / r^2$
C) $k\,p / r^3$
D) Zero

119. Energy stored in 5 μF capacitor at 50 V:
A) 6.25×10^{-3} J
B) 0.625 J
C) 0.125 J
D) 12.5 J

120. Two capacitors 2 μF and 3 μF in series, connected to 10 V. Charge on 2 μF:
A) 12 μC
B) 6 μC
C) 4 μC
D) 10 μC

121. Two capacitors 3 μF and 6 μF in parallel, 12 V. Charge on 6 μF:
A) 72 μC
B) 18 μC
C) 36 μC
D) 6 μC

122. Capacitance of spherical capacitor, inner radius a, outer radius b:
A) $4 \pi \varepsilon_0\, ab / (b - a)$
B) $4 \pi \varepsilon_0\, (b - a) / ab$
C) $4 \pi \varepsilon_0\, (a + b)$
D) $4 \pi \varepsilon_0\, ab$

123. Energy stored in capacitor if dielectric constant K inserted at constant voltage:
A) Increase K times
B) Decrease K times
C) Remains same
D) Zero

124. Electric field inside parallel plate capacitor with dielectric:
A) $E = V/d$
B) $E = V/Kd$
C) $E = KV/d$
D) Zero

125. Work done in assembling 3 charges +q at vertices of equilateral triangle of side a:
A) $3\,k\,q^2 / a$
B) $k\,q^2 / a$
C) $6\,k\,q^2 / a$
D) 0

126. Potential at centroid of equilateral triangle with charges +q at vertices:
A) $k\,q / a$
B) $k\,q / (\sqrt{3}\,a)$
C) 0
D) $k\,q / (3a)$

127. Electric field at midpoint of line joining two equal charges:
A) Zero
B) Maximum
C) $k\,q / r^2$
D) $2\,k\,q / r^2$

128. Force on dipole in non-uniform field:
   A) $F = (p \cdot \nabla)E$
   B) $F = p \times E$
   C) Zero
   D) $F = p E^2$
129. Capacitor 4 μF charged to 100 V, then connected to 6 μF uncharged capacitor. Energy loss:
   A) 40%
   B) 25%
   C) 50%
   D) 10%
130. Two charges +q and -q separated by 2a, potential energy if moved farther apart to 4a:
   A) Decreases by half
   B) Doubles
   C) Remains same
   D) Zero